2 Technical requirements for the SHaPoLi / EPL system

2.1 Required main systems

The SHaPoLi / EPL system should consist of the following main arrangements:

.1 SHaPoLi:

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SOURCE: MEPC.335(76).pdf
Figure 1: Engine load diagram on Shaft/Engine Power Limitation

2 Technical requirements for the SHaPoLi / EPL system

2.1 Required main systems

The SHaPoLi / EPL system should consist of the following main arrangements:

.1 SHaPoLi:

.1 sensors for measuring the torque and rotational speed delivered to the propeller(s) of the ship. The system includes the amplifier and the analogue to the digital converter;

.2 a data recording and processing device for tracking and calculation of the data as given in paragraph 2.2.5.1 of these Guidelines; and

.3 a control unit for calculation and limitation of the power transmitted by the shaft to the propeller(s);

.2 EPL:

.1 for the mechanically controlled engine, a sealing device which can physically lock the fuel index by using a mechanical stop screw sealed by wire or an equivalent device with governor limit setting so that the ship's crew cannot release the EPL without permission from the ship's master or OICNW, as shown in figure 2; or

.2 for the electronically controlled engine, fuel index limiter which can electronically lock the fuel index or direct limitation of the power in

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the ship's master or OICNW, as shown in figure 2; or

.2 for the electronically controlled engine, fuel index limiter which can electronically lock the fuel index or direct limitation of the power in the engine's control system so that the ship's crew cannot release the EPL without permission from the ships master or OICNW; and

.3 where technically possible and feasible, the Sha/PoLi/EPL system should be controlled from the ships bridge and not require attendance in the machinery

space by ship's personnel.

Engine speed
Engine power
Light propeller curve
NMCR 105%NMCR
Available power
Reserved available power
MCR
Maximum
engine speed
ShaPoLi or electronic EPL
Engine speed
Engine power
Light propeller curve
NMCR 105%NMCR
Available power
MCR
Maximum
engine speed
Reserved available power
Mechanical EPL
RESOLUTION MEPC.335(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE SHAFT / ENGINE POWER LIMITATION SYSTEM TO COMPLY WITH THE EEXI REQUIREMENTS
AND USE OF A POWER RESERVEMEPC 76/15/Add.2
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Figure 2: Sealing of mechanical stop screw

2.2 General system requirements

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AND USE OF A POWER RESERVEMEPC 76/15/Add.2
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Figure 2: Sealing of mechanical stop screw

2.2 General system requirements

2.2.1 The SHaPoLi / EPL system should be non- permanent but should require the deliberate action of the ship's master or OICNW to enable the use of unlimited shaft / engine power (power reserve) of the ship. For systems that use a Password/PIN to control access to the power reserve override, attention should be paid to ensure that the necessary Password/PIN is always available when override is required.

2.2.2 For SHaPoLi / EPL system for the electronically control led engine, the control unit should inform the ship's master or OICNW clearly and conspicuously when the ship's shaft /

engine power exceeds the limited shaft / engine power as stated in the Onboard Management Manual (OMM) for SHaPoLi / EPL or in any case of system malfunction.

2.2.3 For EPL for the mechanically controlled engine, the sealing device should either:

.1 visibly indicate removal of the sealing when the ship's engine power exceeds the limited engine power as stated in the OMM for EPL or in any case of

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SOURCE: MEPC.335(76).pdf
2.2.3 For EPL for the mechanically controlled engine, the sealing device should either:

.1 visibly indicate removal of the sealing when the ship's engine power exceeds the limited engine power as stated in the OMM for EPL or in any case of
system malfunction; or

.2 be equipped with other systems such as an alert -monitoring system which can indicate when the ship's engine power exceeds the limited engine power as stated in the OMM for EPL or in any case of system malfuncti on and recording the use of unlimited mode, verified by the Administration or the RO.

2.2.4 The SHaPoLi / EPL system (or each subsystem) should be tamper-proof.

2.2.5 The SHaPoLi / EPL system for the electronically controlled engine should indicate the following data during operation:

.1 for SHaPoLi, shaft rotational speed, shaft torque and shaft power (and total shaft power in case of multiple shaft arrangements) to be recorded constantly in unlimiting mode; or

.2 for EPL, a fuel index sealing system or power limitation system which can indicate and record the use of unlimited mode.

Mechanical Stop Screw

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SOURCE: MEPC.335(76).pdf
in unlimiting mode; or

.2 for EPL, a fuel index sealing system or power limitation system which can indicate and record the use of unlimited mode.

Mechanical Stop Screw
Mechanical stop screw sealed by wire Engine side control console in the governor

2.2.6 The procedure for SHaPoLi / EPL depends on the propulsion system and should be

described in the OMM for SHaPoLi / EPL in accordance with section 4 of these Guidelines.

3 Use of a power reserve by un-limiting the shaft / engine power limitation

3.1 The use of a power reserve is only allowed for the purpose of securing the safety of a ship or saving life at sea, consistent with regulation 3.1 of MARPOL Annex VI (e.g. operating in adverse weather and ice- infested waters, participation in search and rescue operations, avoidance of pirates and engine maintenance). Use of a power reserve should not have

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SOURCE: MEPC.335(76).pdf
in adverse weather and ice- infested waters, participation in search and rescue operations, avoidance of pirates and engine maintenance). Use of a power reserve should not have adverse impact on the propeller, shaft and related systems. It is important that the ship master and OICNW are not restricted from exercising judgement to override the SHaPoLi / EPL when required for safety purposes. The authority for this should be clearly set out in the OMM and/or the Safety Management System manual, as appropriate.

3.2 Any use of a power reserve should be recorded in the record page of the OMM for SHaPoLi / EPL, signed by the master and should be kept on board. The record should include:

.1 ship type;

.2 IMO number;